

泰勒公式 I

$$f = f(x, y), F(t) = f(a + t\Delta x, b + t\Delta y)$$

$$F(t) = \sum_{k=0}^n \frac{1}{k!} F^{(k)}(0) \cdot t^k + \frac{1}{(n+1)!} F^{(n+1)}(\theta t) \cdot t^{n+1} \quad (0 < \theta < 1)$$

$$\begin{aligned} F^{(k)}(t) &= \frac{d^k}{dt^k} f(a + t\Delta x, b + t\Delta y) \\ &= \frac{d^{k-1}}{dt^{k-1}} (\Delta x \cdot \frac{\partial f}{\partial x} + \Delta y \cdot \frac{\partial f}{\partial y}) \quad \text{写作} \quad \frac{d^{k-1}}{dt^{k-1}} (\Delta x \cdot \frac{\partial}{\partial x} + \Delta y \cdot \frac{\partial}{\partial y}) f \end{aligned}$$

而 $\Delta x, \Delta y, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}$ 与 t 无关

$$= (\Delta x \frac{\partial}{\partial x} + \Delta y \frac{\partial}{\partial y})^k f(a + t\Delta x, b + t\Delta y)$$

$$F^{(k)}(0) = (\Delta x \frac{\partial}{\partial x} + \Delta y \frac{\partial}{\partial y})^k f(a, b), \quad F^{(n+1)}(\theta t) = (\Delta x \frac{\partial}{\partial x} + \Delta y \frac{\partial}{\partial y})^{n+1} f(a + \theta t\Delta x, b + \theta t\Delta y)$$

$$\text{令 } t=1, f(a + \Delta x, b + \Delta y) = \sum_{k=0}^n \frac{1}{k!} (\Delta x \frac{\partial}{\partial x} + \Delta y \frac{\partial}{\partial y})^k f(a, b)$$

设 $(a, b) \in \mathbb{R}^2$ 函数 $f = f(x, y)$ 在点 (a, b) 的某邻域 G 内 $n+1$ 阶连续可微, 则对 $(x, y) \in G$, 有

$$f(x, y) = \sum_{k=0}^n \frac{1}{k!} (\Delta x \frac{\partial}{\partial x} + \Delta y \frac{\partial}{\partial y})^k f(a, b) + R_n$$

$$R_n = \frac{1}{(n+1)!} (\Delta x \frac{\partial}{\partial x} + \Delta y \frac{\partial}{\partial y})^{n+1} f(a + \theta \Delta x, b + \theta \Delta y)$$

此为 $f(x, y)$ 在点 (a, b) 处的 n 阶泰勒公式, R_n 为拉格朗日余项

$$\text{若取 } n=0, \text{ 得 } f(x, y) = f(a, b) + f'_x(\xi, \eta)(x-a) + f'_y(\xi, \eta)(y-b)$$

$$\xi = a + \theta(x-a), \eta = b + \theta(y-b)$$

此为二元函数拉格朗日中值公式

若取 $a=b=0$ 得 $f(x,y) = \sum_0^n \frac{1}{k!} \cdot (x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y})^k f(0,0) + R_n$

$$R_n = \frac{1}{(n+1)!} (x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y})^{n+1} \cdot f(\theta x, \theta y) \quad (0 < \theta < 1)$$

此为二元函数的麦克劳林公式

泰勒公式 II

$$f(x,y) = \sum_0^n \frac{1}{k!} (\Delta x \frac{\partial}{\partial x} + \Delta y \frac{\partial}{\partial y})^k f(a,b) + o(\rho^n)$$

$$\Delta x = x - a \quad \Delta y = y - b \quad \rho = \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

此为二元函数在 (a,b) 处带皮亚诺余项的 n 阶泰勒公式